

Supplementary Material

(PRISM: Patient-specific Representation Integration with a Shared Graph Model for Cancer Driver Gene Prioritization)

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Table S1. Summary of the scRNA-seq datasets used in this study, including the GEO accession number, the number of patients, the total number of cells after preprocessing, and the number of patient-specific graphs constructed for each cancer type.

Cancer	GEO Series accession	Patients	Cells	Patient-specific graphs
BRCA	GSE176078	26	100,064	26
BLCA	GSE222315	14	93,090	14
LUAD	GSE123902	13	29,738	13
LIHC	GSE290925	12	23,482	12
LUSC	GSE148071	42	89,887	42
THCA	GSE184362	11	147,204	11
ESCA	GSE160269	60	192,078	60
PRAD	GSE181294	31	487,990	31
STAD	GSE183904	29	127,576	29
COAD	GSE166555	12	38,233	12
CESC	GSE197461	8	35,392	8

Table S2. Dataset statistics across 11 cancer types. Positive samples correspond to driver genes, and negative samples correspond to passenger genes.

	BRCA	BLCA	LUAD	THCA	LIHC	LUSC	ESCA	PRAD	STAD	COAD	CESC
Driver genes	202	95	179	64	82	26	71	50	73	141	17
Passenger genes	2187	2187	2187	2187	2187	2187	2187	2187	2187	2187	2187

Table S3. Comparison of AUPRC (mean $\pm$ standard deviation) on 11 TCGA cancer datasets. The best result for each cancer type is shown in bold.

Cancer	PRISM	EMOGI	MTGCN	DGHNN
BRCA	0.7205 $\pm$ 0.0032	0.6482 $\pm$ 0.0045	0.6583 $\pm$ 0.0040	0.6832 $\pm$ 0.0062
BLCA	0.7237 $\pm$ 0.0040	0.5485 $\pm$ 0.0101	0.6568 $\pm$ 0.0077	0.6935 $\pm$ 0.0064
LUAD	0.6639 $\pm$ 0.0061	0.5591 $\pm$ 0.0105	0.6279 $\pm$ 0.0099	0.6410 $\pm$ 0.0091
LIHC	0.5316 $\pm$ 0.0037	0.3845 $\pm$ 0.0193	0.4645 $\pm$ 0.0195	0.4811 $\pm$ 0.0191
THCA	0.3620 $\pm$ 0.0067	0.2587 $\pm$ 0.0084	0.2907 $\pm$ 0.0110	0.3145 $\pm$ 0.0233
LUSC	0.4685 $\pm$ 0.0224	0.2222 $\pm$ 0.0266	0.3099 $\pm$ 0.0407	0.4054 $\pm$ 0.0542
ESCA	0.4930 $\pm$ 0.0037	0.4174 $\pm$ 0.0297	0.4721 $\pm$ 0.0279	0.4763 $\pm$ 0.0145
PRAD	0.6854 $\pm$ 0.0030	0.5233 $\pm$ 0.0220	0.6143 $\pm$ 0.0189	0.7058 $\pm$ 0.0129
STAD	0.7227 $\pm$ 0.0059	0.4604 $\pm$ 0.0153	0.5931 $\pm$ 0.0117	0.6449 $\pm$ 0.0083
COAD	0.5639 $\pm$ 0.0028	0.3459 $\pm$ 0.0067	0.3816 $\pm$ 0.0085	0.4468 $\pm$ 0.0220
CESC	0.6141 $\pm$ 0.0034	0.5547 $\pm$ 0.0522	0.5972 $\pm$ 0.0528	0.5635 $\pm$ 0.0414

Table S4. Ablation analysis in terms of AUPRC across 11 cancer types. The best result for each cancer type and the highest average are shown in bold.

Cancer	PRISM	PRISM w/o Pretraining	PPI Only	Patient Graphs Only
BRCA	0.7205 $\pm$ 0.0032	0.7141 $\pm$ 0.0089	0.6983 $\pm$ 0.0118	0.5640 $\pm$ 0.0063
BLCA	0.7237 $\pm$ 0.0040	0.7271 $\pm$ 0.0083	0.7097 $\pm$ 0.0190	0.6449 $\pm$ 0.0130
LUAD	0.6639 $\pm$ 0.0061	0.6574 $\pm$ 0.0057	0.6392 $\pm$ 0.0181	0.5708 $\pm$ 0.0066
LIHC	0.5316 $\pm$ 0.0037	0.5208 $\pm$ 0.0184	0.4578 $\pm$ 0.0438	0.4883 $\pm$ 0.0130
THCA	0.3620 $\pm$ 0.0067	0.3147 $\pm$ 0.0290	0.3031 $\pm$ 0.0175	0.2009 $\pm$ 0.0127
LUSC	0.4685 $\pm$ 0.0224	0.4258 $\pm$ 0.0294	0.3744 $\pm$ 0.0504	0.3496 $\pm$ 0.0267
ESCA	0.4930 $\pm$ 0.0037	0.4901 $\pm$ 0.0197	0.4382 $\pm$ 0.0321	0.4561 $\pm$ 0.0129
PRAD	0.6854 $\pm$ 0.0030	0.6834 $\pm$ 0.0117	0.6783 $\pm$ 0.0325	0.5007 $\pm$ 0.0221
STAD	0.7227 $\pm$ 0.0059	0.7087 $\pm$ 0.0090	0.6713 $\pm$ 0.0278	0.6372 $\pm$ 0.0244
COAD	0.5639 $\pm$ 0.0028	0.5425 $\pm$ 0.0256	0.5247 $\pm$ 0.0238	0.5059 $\pm$ 0.0049
CESC	0.6141 $\pm$ 0.0034	0.5060 $\pm$ 0.0238	0.4761 $\pm$ 0.0445	0.4099 $\pm$ 0.0365